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Part 1: Asphalt Concrete

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English Version

Bituminous mixtures - Material specifications - Part 1: Asphalt Concrete

Mélanges bitumineux - Spécifications sur le matériau -
Partie 1: Enrobés bitumineux

Asphaltmischgut - Mischgutanforderungen - Teil 1:
Asphaltbeton

This draft European Standard is submitted to CEN members for enquiry. It has been drawn up by the Technical Committee CEN/TC 227.

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Foreword

This document (prEN 13108-1:2013) has been prepared by Technical Committee CEN/TC 227 “Road materials”, the secretariat of which is held by DIN.

This document is currently submitted to the CEN Enquiry.

This document will supersede EN 13108-1:2006.

This document has been prepared under a mandate given to CEN by the European Commission and the European Free Trade Association, and supports essential requirements of EU Directive(s).

For relationship with EU Directive(s), see informative Annex ZA which is an integral part of this document.

This European Standard is one of a series of standards as listed below:

prEN 13108-1, *Bituminous mixtures — Material specifications — Part 1: Asphalt Concrete.*

prEN 13108-2, *Bituminous mixtures — Material specifications — Part 2: Very Thin Layer Asphalt Concrete (BBTM).*

prEN 13108-3, *Bituminous mixtures — Material specifications — Part 3: Soft Asphalt.*

prEN 13108-4, *Bituminous mixtures — Material specifications — Part 4: Hot Rolled Asphalt.*

prEN 13108-5, *Bituminous mixtures — Material specifications — Part 5: Stone Mastic Asphalt.*

prEN 13108-6, *Bituminous mixtures — Material specifications — Part 6: Mastic Asphalt.*

prEN 13108-7, *Bituminous mixtures — Material specifications — Part 7: Porous Asphalt.*

prEN 13108-8, *Bituminous mixtures — Material specifications — Part 8: Reclaimed Asphalt.*

prEN 13108-9, *Bituminous mixtures — Material specifications — Part 9: Asphalt for Ultra-Thin Layer (AUTL).*

prEN 13108-20, *Bituminous mixtures — Material specifications — Part 20: Type Testing.*

prEN 13108-21, *Bituminous mixtures — Material specifications — Part 21: Factory Production Control.*

Annex A (normative) details the calculation of the penetration or the softening point in mixtures containing reclaimed asphalt from the penetrations or softening points of the added binder and the recovered binder from the reclaimed asphalt.

Annex B (informative) gives information on the use of the proposed test method for friction after polishing

NOTE Standards for mix groups utilizing cutback bitumen or bitumen emulsion and for materials based on in situ recycling will be considered by CEN/TC 227 when the hot mix asphalt standards have been completed.

Introduction

The ultimate aim is to specify bituminous mixtures on a performance level. For asphalt concrete besides traditional (empirical) tests more tests are available with a more direct relation to the performance in the road. Although more tests are available to provide information on the performance of the material, they are still insufficient to cover the actual performance of the material in the road. Attention should be paid to possible conflicting specifications.

Asphalt Concrete is to be used for surface courses, binder courses, regulating courses, and bases.

1 Scope

This European Standard specifies requirements for mixtures of the mix group Asphalt Concrete for use on roads, airfields and other trafficked areas.

This European Standard includes requirements for the selection of the constituent materials. It is designed to be read in conjunction with prEN 13108-20 and prEN 13108-21

2 Normative references

The following documents, in whole or in part, are normatively referenced in this document and are indispensable for its application. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

EN 1097-6, *Tests for mechanical and physical properties of aggregates — Part 6: Determination of particle density and water absorption*

EN 1426, *Bitumen and bituminous binders — Determination of needle penetration*

EN 1427, *Bitumen and bituminous binders — Determination of the softening point — Ring and ball method*

EN 12591, *Bitumen and bituminous binders — Specifications for paving grade bitumens*

EN 12697-3, *Bituminous mixtures — Test methods for hot mix asphalt — Part 3: Bitumen recovery: Rotary evaporator*

EN 12697-4, *Bituminous mixtures — Test methods for hot mix asphalt — Part 4: Bitumen recovery: Fractionating column*

EN 12697-8, *Bituminous mixtures — Test methods for hot mix asphalt — Part 8: Determination of void characteristics of bituminous specimens*

EN 12697-12, *Bituminous mixtures — Test methods for hot mix asphalt — Part 12: Determination of the water sensitivity of bituminous specimens*

EN 12697-13, *Bituminous mixtures — Test methods for hot mix asphalt — Part 13: Temperature measurement*

EN 12697-16, *Bituminous mixtures — Test methods for hot mix asphalt — Part 16: Abrasion by studded tyres*

EN 12697-22, *Bituminous mixtures — Test methods for hot mix asphalt — Part 22: Wheel tracking*

EN 12697-24, *Bituminous mixtures — Test methods for hot mix asphalt — Part 24: Resistance to fatigue*

EN 12697-25, *Bituminous mixtures — Test methods for hot mix asphalt — Part 25: Cyclic compression test*

EN 12697-26, *Bituminous mixtures — Test methods for hot mix asphalt — Part 26: Stiffness*

EN 12697-31, *Bituminous mixtures — Test methods for hot mix asphalt — Part 31: Specimen preparation by gyratory compactor*

EN 12697-34, *Bituminous mixtures — Test methods for hot mix asphalt — Part 34: Marshall test*

EN 12697-41, *Bituminous mixtures — Test methods for hot mix asphalt — Part 41: Resistance to de-icing fluids*

EN 12697-43, *Bituminous mixtures — Test methods for hot mix asphalt — Part 43: Resistance to fuel*

EN 12697-44, *Bituminous mixtures — Test methods for hot mix asphalt — Part 44: Crack propagation by semi-circular bending test*

EN 12697-45, *Bituminous mixtures — Test methods for hot mix asphalt — Part 45: Saturation Ageing Tensile Stiffness (SATS) conditioning test*

EN 12697-46, *Bituminous mixtures — Test methods for hot mix asphalt — Part 46: Low temperature cracking and properties by uniaxial tension tests*

EN 13043, *Aggregates for bituminous mixtures and surface treatments for roads, airfields and other trafficked areas*

prEN 13108-4:2013, *Bituminous mixtures — Material specifications — Part 4: Hot Rolled Asphalt*

prEN 13108-8, *Bituminous mixtures — Material specifications — Part 8: Reclaimed Asphalt*

prEN 13108-20:2013, *Bituminous mixtures — Material specifications — Part 20: Type Testing*

prEN 13108-21:2013, *Bituminous mixtures — Material specifications — Part 21: Factory Production Control*

EN 13501-1, *Fire classification of construction products and building elements — Part 1: Classification using data from reaction to fire tests*

EN 13924, *Bitumen and bituminous binders — Specifications for hard paving grade bitumens*

EN 14023, *Bitumen and bituminous binders — Specification framework for polymer modified bitumens*

EN ISO 11925-2, *Reaction to fire tests — Ignitability of products subjected to direct impingement of flame — Part 2: Single-flame source test (ISO 11925-2)*

3 Terms, definitions, symbols and abbreviations

3.1 Terms and definitions

For the purposes of this document, the following terms and definitions apply.

3.1.1

pavement

structure, composed of one or more courses, to assist the passage of traffic over terrain

3.1.2

layer

element of a pavement laid in a single operation

3.1.3

course

structural element of a pavement constructed with a single material. A course may be laid in one or more layers

3.1.4

surface course

upper course of the pavement, which is in contact with the traffic

3.1.5

binder course

part of the pavement between the surface course and the base

3.1.6

regulating course

course of variable thickness applied to an existing course or surface to provide the necessary profile for a further course of consistent thickness

3.1.7

base

main structural element of a pavement. The base may be laid in one or more courses, described as “upper” base, “lower” base

3.1.8

Asphalt

homogenous mixture of coarse and fine aggregates, filler aggregate and bituminous binder which is used in the construction of flexible pavement layers

Note 1 to entry: The asphalt may include one or more additives to enhance the laying characteristics, performance or appearance of the mixture. Natural asphalt is defined in prEN 13108-4.

3.1.9

Asphalt Concrete

asphalt in which the aggregate particles are continuously graded or gap-graded to form an interlocking structure

3.1.10

mix formulation

composition of a single mixture expressed as a target composition

Note 1 to entry: A target composition may be expressed in two ways (see 3.1.11 and 3.1.12).

3.1.11

input target composition

expression of a mix formulation in terms of the constituent materials, the grading curve and the percentage of bitumen added to the mixture

Note 1 to entry: This will usually be the result of a laboratory mix design and validation.

3.1.12

output target composition

expression of a mix formulation in terms of the constituent materials and the mid point grading and soluble binder content to be found on analysis

Note 1 to entry: This will usually be the result of a production validation.

3.1.13

additive

constituent material, which can be added in small quantities to the mixture, e. g. inorganic or organic fibres or polymers, to influence the mechanical properties, the workability or the colour of the mixture

3.1.14

conflicting specification

combination of requirements or properties which are impossible to fulfil in their entirety

Note 1 to entry: This can occur by combining specific requirements for the composition and constituent materials together with more performance related tests, or when two or more performance or test parameters are selected which measure similar properties using conflicting test methods resulting in a lack of clarity and consistency in the characteristics of the mixture.

3.2 Symbols and abbreviations

AC general designation of Asphalt Concrete

AC *D* designation of Asphalt Concrete followed by an indication of *D*, the upper sieve size of the aggregate in the mixture, in millimetres (mm)

EXAMPLE AC 6 Asphalt Concrete with an upper sieve size of the aggregate of 6 mm.

4 Requirements for constituent materials

4.1 General

Only constituent materials with established suitability shall be used. For all constituent materials the relevant properties shall be made available

The establishment of suitability shall result from one or more of the following:

- European Standard;
- European Technical Assessment;
- specifications for materials based on a demonstrable history of satisfactory use in asphalt. Evidence shall be provided on their suitability. This evidence may be based on research combined with evidence from practice.

NOTE 1 In the European asphalt industry it is practice to use constituent materials which are not covered by a European Standard or European Technical Assessment.

NOTE 2 Chemical and organic additives may be used to reduce production temperatures by influencing the viscosity of the binder. This can have an effect on the rheology of binders at temperatures in the pavement thereby influencing relevant properties. Attention should be given to the choice of requirements to determine the effect of such an additive on those properties.

NOTE 3 Attention should be given to the effects of all kinds of constituent materials on the potential for future recycling.

4.2 Binder

4.2.1 General

The binder shall be a paving grade bitumen, a modified bitumen, a hard grade bitumen or a blend of either with natural asphalt. The paving grade bitumen shall conform to EN 12591, the modified bitumen to EN 14023 and the hard grade bitumen to EN 13924. When natural asphalt is used, it shall conform to prEN 13108-4:2013, Annex B.

The grade of the bitumen, the type and grade of modified bitumen and the amount and category of natural asphalt according to prEN 13108-4:2013, Annex B, shall be declared.

Premix bituminous binders and other bituminous binders that are not covered by EN 12591 or EN 14023 can be used provided that information is given as stated in 4.1, and that the base bitumen is conforming to EN 12591, EN 14023 or EN 13924.

NOTE 1 It is practice in Europe for special purposes to use other binders or premixed binders (binders mixed with additives or polymers in the asphalt plant).

NOTE 2 Asphalt mixes with chemical modified binders not covered by EN 14023 are not covered by this European Standard.

4.2.2 Selection of binder

Depending on the conditions of use, the grade of the bitumen, the type and grade of modified bitumen and the amount and category of natural asphalt may be specified.

NOTE 1 This European Standard allows for a combination of requirements. The user of the European Standard should take good care of the choice and combination of requirements. Not all requirements or combinations of requirements give adequate information on the expected performance of the material. In order to determine the influence of special binders (PMB's) or additives it is necessary to choose requirements that have an expected relationship with the performance.

NOTE 2 Traditionally the grades of a paving grade bitumen are selected from the grades between 20/30 and 330/430 inclusively and in case of a hard grade bitumen the grade is selected from the grades 10/20 and 15/25.

NOTE 3 EN 14023 for modified bitumen is a grading system and is only meant to characterise the modified bitumen. The modified bitumen specifications are not performance based. This means that not all requirements or combinations of requirements give information on the expected behaviour of the asphalt mix.

When an additive is used to lower the production temperature of the asphalt concrete and thereby changes relevant properties of the binder at ambient temperatures evidence shall be provided to show what this influence of the additive is on the performance of the mix. This evidence may be based on earlier research.

4.2.3 Surface courses with reclaimed asphalt

When using more than 10 % by mass of the total mixture of reclaimed asphalt from mixtures in which only paving grade bitumen has been used and when the binder added to the mixture is a paving grade bitumen and the grade of the bitumen is selected, the following requirements may be specified.

The penetration or the softening point of the binder in the resulting mixture, calculated from the penetrations or the softening points of the added binder and the recovered binder from the reclaimed asphalt, shall meet the penetration or softening point requirements of the specified grade. The calculation shall be executed according to Annex A. Either the penetration or the softening point requirement shall be selected.

When using reclaimed asphalt from mixtures in which a modified bitumen and/or a modifier additive has been used, and/or the mixture itself contains a modified bitumen or a modifier, the amount of reclaimed asphalt may be limited to a maximum of 10 % by mass of the total mixture.

NOTE 1 The choice for this specification depends on the choice of requirements with in this European Standard. For more performance designed mixes there may be no need to apply the pen or softening point rule. It should however be noted that the pen or softening point rule is only valid for unmodified bitumen).

NOTE 2 When applying an empirical approach to the mixture using a too great a proportion of modified bitumen or modifier could lead to an incorrect decision in respect to the addition of the new binder.

The range of acceptable types and grades of bitumen which may be used may be specified.

NOTE 3 In some cases the binder of the recovered asphalt can be so hardened that a very soft bitumen has to be chosen to fulfil the requirement in 4.2.3. In such cases it may be appropriate to choose an alternative grade to that calculated according Annex A (normative). This may be specified on the basis of experience in the place of use.

4.2.4 Regulating courses, binder courses and bases with reclaimed asphalt

When using more than 20 % by mass of the total mixture of reclaimed asphalt from mixtures in which only paving grade bitumen has been used and when the binder added to the mixture is a paving grade bitumen and the grade of the bitumen is selected, the following requirement may be specified.

The penetration or the softening point of the binder in the resulting mixture, calculated from the penetrations or the softening points of the added binder and the recovered binder from the reclaimed asphalt, shall meet the penetration or softening point requirements of the selected grade. The calculation shall be executed according to Annex A (normative). Either the penetration or the softening point requirement shall be selected.

When using reclaimed asphalt from mixtures in which a modified bitumen and/or a modifier additive has been used, and/or the mixture itself contains a modified bitumen or a modifier, the amount of reclaimed asphalt for regulating courses, binder courses and base courses may be limited to a maximum of 20 % by mass of the total mixture.

NOTE 1 The choice for this specification depends on the choice of requirements within this standard. For more performance designed mixes there may be no need to apply the pen or softening point rule. (It should however be noted that the pen or softening point rule is only valid for unmodified bitumen).

NOTE 2 When applying an empirical approach to the mixture using a too great a proportion of modified bitumen or modifier could lead to an incorrect decision in respect to the addition of the new binder.

The range of acceptable types and grades of bitumen which may be used may be specified.

NOTE 3 In some cases the binder of the recovered asphalt can be so hardened that a very soft bitumen has to be chosen to fulfil the requirement in 4.2.3. In such cases it may be appropriate to choose an alternative grade to that calculated according Annex A (normative). This may be specified on the basis of experience in the place of use.

4.3 Aggregates

NOTE The expression “as appropriate for the intended use” in 4.3.1 to 4.3.4 means that the selection of the requirements and the particular category depends on a number of conditions. These conditions will include traffic density, climatic conditions, the construction of the course in which the mixture will be used, and economic considerations.

4.3.1 Coarse aggregate

Coarse aggregate shall conform to EN 13043 as appropriate for the intended use.

4.3.2 Fine aggregate

Fine aggregate shall conform to EN 13043 as appropriate for the intended use.

4.3.3 All-in aggregates

All-in aggregate shall conform to EN 13043 as appropriate for the intended use.

4.3.4 Added filler

Filler aggregate shall conform to EN 13043 as appropriate for the intended use. Based on the experience in the place of use the amount of added filler may be specified.

NOTE Filler includes materials such as cement and hydrated lime.

4.4 Reclaimed asphalt

NOTE The expression “appropriate to the intended use” means that the selection of the requirements and the particular category depends on a number of conditions. These conditions will include traffic density, climatic conditions, the construction of the course in which the mixture will be used, and economic considerations.

The use and the amount of reclaimed asphalt, and the mix group and/or the courses from which the reclaimed asphalt has been or will be derived may be specified.

The properties of reclaimed asphalt declared in accordance with prEN 13108-8 shall conform to requirements that may be selected appropriate to the intended use according to of prEN 13108-8:2013, 4.2 and 4.3.

The upper sieve size D of the aggregate in the reclaimed asphalt shall not exceed the upper sieve size D of the mixture. The aggregate properties of the reclaimed asphalt or of the mixed aggregates from the reclaimed asphalt with the virgin aggregates shall fulfil the requirements specified for the aggregate for the mixture.

The amount of reclaimed asphalt in the mixture and when required the mix group and /or the courses from which the reclaimed asphalt has been or will be derived shall be declared.

4.5 Additives

The nature and properties of all additives shall be declared and they shall conform to the requirements referred to in 4.1. For specific applications and based of the experience in the place of use the amount of additives may be specified.

5 Requirements for the mixture

5.1 General

The target composition of the mixture in terms of its constituent materials, the percentages passing the specified sieves, the target binder content and where relevant the binder from reclaimed asphalt and/or natural asphalt and the percentage(s) of additive(s) shall be declared and documented. The target grading curve shall be declared for the sieve 1,4 *D* and the sieves as called up in Table 3.

At the target composition the mixture shall conform to the specified requirements in accordance with this European Standard.

The test results in accordance with prEN 13108-20: 2013, 7.5, shall be made available

5.2 Composition, grading, binder content

5.2.1 Composition

The grading shall be expressed in percentages by mass of total aggregate. The binder and additive content shall be expressed in percentages by mass of the total mixture. The percentages passing the sieves, with exception of the sieve 0,063 mm shall be expressed to 1 %, the binder content, the percentage passing sieve 0,063 mm and any additive content shall be expressed to 0,1 %. Where appropriate the additive content shall be expressed to 0,01 %.

NOTE Sometimes an additive is added in very small proportions. In that case it is needed to express the amount to 0,01 %.

5.2.2 Grading

5.2.2.1 Overall grading requirements

The requirements for the overall grading limits are given in Tables 1 and 2 for the sieves 1,4 *D*, *D*, 2 mm and 0,063 mm. The target composition shall be within this grading envelope (maximum and minimum values included).

The sieves to be used shall be either basic sieve set plus set 1 or basic sieve set plus set 2, according to EN 13043. A combination of sieve sizes from set 1 and set 2 shall not be permissible.

Table 1 — Overall limits of target composition — basic sieve set plus set 1

<i>D</i>	4	5 (5,6)	8	11 (11,2)	16	22 (22,4)	32 (31,5)
Sieve mm	Percentage passing by mass						
1,4 <i>D</i> ^a	100	100	100	100	100	100	100
<i>D</i>	90 to 100	90 to 100	90 to 100	90 to 100	90 to 100	90 to 100	90 to 100
2	50 to 85	15 to 72	10 to 72	10 to 60	10 to 50 ^b	10 to 50 ^b	10 to 65
0,063	5,0 to 17,0	2,0 to 15,0	2,0 to 13,0	2,0 to 12,0	0 to 12,0	0 to 11,0	0 to 11,0
^a Where the sieve calculated as 1,4 <i>D</i> is not an exact number in the basic set plus set 1 series then the next nearest sieve in the set shall be adopted. ^b For application on airfields the maximum percentage passing 2 mm may be increased to 60 %.							

Table 2 — Overall limits of target composition — basic sieve set plus set 2

<i>D</i>	4	6 (6,3)	8	10	12 (12,5)	14	16	20	32 (31,5)
Sieve mm	Percentage passing by mass								
1,4 <i>D</i> ^a	100	100	100	100	100	100	100	100	100
<i>D</i>	90 to 100	90 to 100	90 to 100	90 to 100	90 to 100	90 to 100	90 to 100	90 to 100	90 to 100
2	50 to 85	15 to 72	10 to 72	10 to 60	10 to 55	10 to 50 ^b	10 to 50 ^b	10 to 50 ^b	10 to 65
0,063	5,0 to 17,0	2,0 to 15,0	2,0 to 13,0	2,0 to 12,0	2,0 to 12,0	0 to 12,0	0 to 12,0	0 to 11,0	0 to 11,0
^a Where the sieve calculated as 1,4 <i>D</i> is not an exact number in the basic set plus set 2 series then the next nearest sieve in the set shall be adopted. ^b For application on airfields the maximum percentage passing 2 mm may be increased to 60 %.									

5.2.2.2 Grading envelope

When required, the grading requirements for the target composition shall be expressed in terms of a grading envelope by selection of maximum and minimum values for the percentages passing the sieves 1,4 *D*; *D*; a characteristic coarse sieve, 2 mm, a characteristic fine sieve and 0,063 mm.

D and the sieves between *D* and 2 mm shall be selected from the following sieves:

- basic sieve set plus set 1: 4 mm; 5,6 mm; 8 mm; 11,2 mm; 16 mm; 22,4 mm, 31,5 mm;
- basic sieve set plus set 2: 4 mm; 6,3 mm; 8 mm; 10 mm; 12,5 mm; 14 mm; 16 mm; 20 mm, 31,5 mm.

In addition the requirements for the grading envelope, when required, may include the percentages passing one optional sieve between *D* and 2 mm and one optional fine sieve between 2 mm and 0,063 mm.

The fine sieves shall be selected from the following sieves: 1 mm; 0,5 mm; 0,25 mm and 0,125 mm.

The target composition of the mix shall be within this grading envelope.(maximum and minimum values included). The ranges between the maximum and minimum values for the grading envelope shall be selected as a single value within the given limits (both included) from Table 3. The percentage passing the sieves *D*, 2 mm and 0,063 mm, of the selected grading envelope shall not exceed the maximum and minimum values in Table 1 or Table 2.

Table 3 — Ranges between maximum and minimum values for the selected grading envelope

Sieve mm	Ranges for grading envelope
<i>D</i>	10
Characteristic coarse sieve	10 to 30 ^a
Optional extra sieve between <i>D</i> and 2	10 to 30 ^a
2	5 to 25 ^a
Characteristic fine	4 to 25 ^a
Optional extra sieve between 2 and 0,063	4 to 25 ^a
0,063	2 to 12 ^a
^a Value to be selected within the given minimum and maximum value (both included).	

In the case of a fixed target grading meeting the requirements of Table 1 or Table 2 the use of the minimum range values in Table 3 is not required.

NOTE Tables 1 and 2 give the overall requirements for the grading of asphalt concrete. Depending on the experience with the combination of requirements in this standard more or less degrees of freedom for the producer may be given for the composition and grading of asphalt concrete. Using the more performance characteristics (stiffness, fatigue, a.o) opens the possibility to give the manufacturer all degrees of freedom within the overall limits of the target composition to choose his grading. With less experience and a more traditional recipe based approach there is a need to define the grading more precise and a requirement for characteristic and optional sieves might be necessary. Depending on the experience in the place of use and in accordance with this specification the ranges between the minimum and maximum values of the grading envelope can be chosen.

5.2.3 Binder content

The requirement for the minimum binder content for the target composition shall be selected from the categories in Table 4. The binder content requirement shall be corrected by multiplying by the factor

$$\alpha = \frac{2,650}{\rho_a}$$

where

ρ_a is the apparent particle dry density of the aggregate, in megagrams per cubic metre (Mg/m³), determined on the weighted mean of the total mineral fraction according to EN 1097-6.

Based on experience in the place of use and for certain specific aggregates the corrected minimum binder content may be adjusted. The adjustment shall be documented and declared.

For porous aggregates the density will be calculated according EN 12697-5 (procedure A). The apparent density shall be declared.

NOTE The binder content includes that contained in reclaimed asphalt and natural asphalt when used.

Table 4 — Minimum binder content, $B_{\min}$

Minimum binder content %	Category $B_{\min}$
3,0	$B_{\min 3,0}$
3,2	$B_{\min 3,2}$
3,4	$B_{\min 3,4}$
3,6	$B_{\min 3,6}$
3,8	$B_{\min 3,8}$
4,0	$B_{\min 4,0}$
4,2	$B_{\min 4,2}$
4,4	$B_{\min 4,4}$
4,6	$B_{\min 4,6}$
4,8	$B_{\min 4,8}$
5,0	$B_{\min 5,0}$
5,2	$B_{\min 5,2}$
5,4	$B_{\min 5,4}$
5,6	$B_{\min 5,6}$
5,8	$B_{\min 5,8}$
6,0	$B_{\min 6,0}$
6,2	$B_{\min 6,2}$
6,4	$B_{\min 6,4}$
6,6	$B_{\min 6,6}$
6,8	$B_{\min 6,8}$
7,0	$B_{\min 7,0}$
7,2	$B_{\min 7,2}$
7,4	$B_{\min 7,4}$
7,6	$B_{\min 7,6}$
7,8	$B_{\min 7,8}$
8,0	$B_{\min 8,0}$
NR	$B_{\min \text{ NR}}$

5.3 Properties

5.3.1 Specimens

For application of this European Standard specimens shall be in accordance with prEN 13108-20:2013, 6.5.

5.3.2 Void content requirements

5.3.2.1 Void content

The void content shall be determined in accordance with EN 12697-8 using the conditions defined in prEN 13108-20:2013, D.2.

The void content shall comply with maximum and minimum categories selected from Tables 5 and 6.

The compaction of test specimens shall be selected from prEN 13108-20:2013, Table C.1.

Table 5 — Maximum void content, $V_{\max}$

Maximum void content %	Category $V_{\max}$
2,0	$V_{\max 2,0}$
2,5	$V_{\max 2,5}$
3,0	$V_{\max 3,0}$
3,5	$V_{\max 3,5}$
4,0	$V_{\max 4,0}$
4,5	$V_{\max 4,5}$
5,0	$V_{\max 5,0}$
5,5	$V_{\max 5,5}$
6,0	$V_{\max 6,0}$
7,0	$V_{\max 7,0}$
8,0	$V_{\max 8,0}$
9,0	$V_{\max 9,0}$
10,0	$V_{\max 10,0}$
11,0	$V_{\max 11,0}$
12,0	$V_{\max 12,0}$
13,0	$V_{\max 13,0}$
14,0	$V_{\max 14,0}$
No requirement	$V_{\max \text{ NR}}$

Table 6 — Minimum void content, $V_{\min}$

Minimum void content %	Category $V_{\min}$
0,5	$V_{\min 0,5}$
1,0	$V_{\min 1,0}$
1,5	$V_{\min 1,5}$
2,0	$V_{\min 2,0}$
2,5	$V_{\min 2,5}$
3,0	$V_{\min 3,0}$
3,5	$V_{\min 3,5}$
4,0	$V_{\min 4,0}$
4,5	$V_{\min 4,5}$
5,0	$V_{\min 5,0}$
5,5	$V_{\min 5,5}$
6,0	$V_{\min 6,0}$
No requirement	$V_{\min NR}$

5.3.2.2 Voids filled with bitumen

The percentage of voids filled with bitumen shall be determined according EN 12697-8 using the conditions defined in prEN 13108-20:2013, D.2 based on the samples prepared and tested according to 5.3.2.1.

The minimum and maximum percentage of voids filled with bitumen shall comply with categories selected from Tables 7 and 8.

Table 7 — Minimum voids filled with bitumen, $VFB_{\min}$

Minimum voids filled with bitumen %	Category $VFB_{\min}$
50	$VFB_{\min 50}$
55	$VFB_{\min 55}$
60	$VFB_{\min 60}$
65	$VFB_{\min 65}$
70	$VFB_{\min 70}$
72	$VFB_{\min 72}$
75	$VFB_{\min 75}$
78	$VFB_{\min 78}$
No requirement	$VFB_{\min NR}$

Table 8 — Maximum voids filled with bitumen, $VFB_{\max}$

Maximum voids filled with bitumen %	Category $VFB_{\max}$
50	$VFB_{\max}$ 50
53	$VFB_{\max}$ 53
56	$VFB_{\max}$ 56
59	$VFB_{\max}$ 59
62	$VFB_{\max}$ 62
65	$VFB_{\max}$ 65
68	$VFB_{\max}$ 68
71	$VFB_{\max}$ 71
74	$VFB_{\max}$ 74
77	$VFB_{\max}$ 77
80	$VFB_{\max}$ 80
83	$VFB_{\max}$ 83
86	$VFB_{\max}$ 86
89	$VFB_{\max}$ 89
93	$VFB_{\max}$ 93
97	$VFB_{\max}$ 97
No requirement	$VFB_{\max}$ NR

5.3.2.3 Voids in mineral aggregate

The percentage of voids in mineral aggregate shall be determined according prEN 12697-8 using the conditions defined in prEN 13108-20:2013, D.2 based on the samples prepared and tested according to 5.3.2.1.

The minimum percentage of voids in mineral aggregate shall comply with the category selected from Table 9.

Table 9 — Minimum voids in mineral aggregate, $VMA_{\min}$

Minimum voids in mineral aggregate %	Category $VMA_{\min}$
8	$VMA_{\min 8}$
10	$VMA_{\min 10}$
12	$VMA_{\min 12}$
14	$VMA_{\min 14}$
16	$VMA_{\min 16}$
18	$VMA_{\min 18}$
No requirement	$VMA_{\min \text{NR}}$

5.3.2.4 Void content at 10 gyrations

The minimum void content after 10 gyrations shall be determined in accordance with EN 12697-31 using the conditions defined in prEN 13108-20:2013, D.2.

The minimum void content after compaction by 10 gyrations shall comply with the category selected from Table 10.

Table 10 — Minimum void content at 10 gyrations, $V10G_{\min}$

Minimum void content at 10 gyrations %	Category $V10G_{\min}$
14,0	$V10G_{\min 14,0}$
11,0	$V10G_{\min 11,0}$
9,0	$V10G_{\min 9,0}$
No requirement	$V10G_{\min \text{NR}}$

5.3.3 Water sensitivity

The water sensitivity shall be determined in accordance with EN 12697-12 using the conditions defined in prEN 13108-20:2013, D.3.

The water sensitivity of specimens shall comply with the category selected from Table 11.

Table 11 — Minimum Water-sensitivity, $ITSR_A$ or i/C_B

Minimum Water-sensitivity %	Category	
	Method A	Method B
90	$ITSR_{90}$	i/C_{90}
80	$ITSR_{80}$	i/C_{80}
70	$ITSR_{70}$	i/C_{70}
60	$ITSR_{60}$	i/C_{60}
No requirement	$ITSR_{NR}$	i/C_{NR}

5.3.4 Resistance to abrasion by studded tyres

The resistance to abrasion by studded tyres shall be determined in accordance with EN 12697-16 using the conditions defined in prEN 13108-20: 2013, D.4.

The resistance to abrasion by studded tyres of specimens shall comply with the category selected from Table 12.

The compaction of test specimens shall be selected from prEN 13108-20:2013, Table C.1.

The range between the upper and lower limits selected shall be 2 % based on compaction degree and 3 % based on void content.

Table 12 — Maximum abrasion value, Abr_A

Maximum abrasion value ml	Abr_A
20	$Abr_{A\ 20}$
24	$Abr_{A\ 24}$
28	$Abr_{A\ 28}$
32	$Abr_{A\ 32}$
36	$Abr_{A\ 36}$
40	$Abr_{A\ 40}$
45	$Abr_{A\ 45}$
50	$Abr_{A\ 50}$
No requirement	Abr_{NR}

NOTE According to Nordic experiences the correlation between laboratory and abrasion in field is not established when polymer modified bitumen or rubber (or a similar binder) is used. The reason is the elastic effect that these binders have during testing. Special care should be taken when specifying requirements on Prall values.

5.3.5 Resistance to permanent deformation

5.3.5.1 Rut depth and rut rate in wheel-tracking

The resistance to permanent deformation, in terms of wheel tracking, shall be determined in accordance with EN 12697-22 using the conditions defined in prEN 13108-20:2013, D.6.

The resistance to permanent deformation of specimens shall comply with the categories selected from one of the Tables 13, 14, 15 or 16. A combination of a requirement from Tables 15 and 16 is not allowed.

The compaction of test specimens shall be selected from prEN 13108-20:2013, Table C.1.

The range between the upper and lower limits selected shall be 2 % based on compaction degree and 3 % based on void content.

The void content of the specimens shall be specified in accordance with prEN 13108-20:2013, D.1.

Table 13 — Resistance to permanent deformation, large size device^a, maximum proportional rut depth, P

Maximum proportional rut depth %	Category P
5,0	$P_{5,0}$
7,5	$P_{7,5}$
10,0	$P_{10,0}$
15,0	$P_{15,0}$
20,0	$P_{20,0}$
No requirement	P_{NR}
^a For designed axle loads ≥ 13 Mg.	

Table 14 — Resistance to permanent deformation, small size device^a, procedure B, conditioning in air, maximum wheel tracking slope, WTS_{AIR}

Maximum wheel tracking slope mm per 10^3 load cycle	Category WTS_{AIR}
0,03	$WTS_{AIR\ 0,03}$
0,05	$WTS_{AIR\ 0,05}$
0,07	$WTS_{AIR\ 0,07}$
0,10	$WTS_{AIR\ 0,10}$
0,15	$WTS_{AIR\ 0,15}$
0,30	$WTS_{AIR\ 0,30}$
0,40	$WTS_{AIR\ 0,40}$
0,50	$WTS_{AIR\ 0,50}$
0,60	$WTS_{AIR\ 0,60}$
0,80	$WTS_{AIR\ 0,80}$
1,00	$WTS_{AIR\ 1,00}$
No requirement	$WTS_{AIR\ NR}$
^a For designed axle loads < 13 Mg.	

Table 15 — Resistance to permanent deformation, small size device^a, procedure B, conditioning in air, maximum proportional rut depth, PR_{DAIR}

Maximum proportional rut depth %	Category PR_{DAIR}
3,0	$PR_{DAIR\ 3,0}$
5,0	$PR_{DAIR\ 5,0}$
7,0	$PR_{DAIR\ 7,0}$
9,0	$PR_{DAIR\ 9,0}$
11,0	$PR_{DAIR\ 11,0}$
13,0	$PR_{DAIR\ 13,0}$
16,0	$PR_{DAIR\ 16,0}$
No requirement	$PR_{DAIR\ NR}$
^a For designed axle loads < 13 Mg.	

Table 16 — Resistance to permanent deformation, small size device^a, procedure B, conditioning in air, maximum rut depth, RD_{AIR}

Maximum rut depth mm	Category RD_{AIR}
1,0	$RD_{AIR\ 1,0}$
1,5	$RD_{AIR\ 1,5}$
2,0	$RD_{AIR\ 2,0}$
2,5	$RD_{AIR\ 2,5}$
3,0	$RD_{AIR\ 3,0}$
3,5	$RD_{AIR\ 3,5}$
4,0	$RD_{AIR\ 4,0}$
4,5	$RD_{AIR\ 4,5}$
5,0	$RD_{AIR\ 5,0}$
No requirement	$RD_{AIR\ NR}$
^a For designed axle loads < 13 Mg.	

5.3.5.2 Resistance to permanent deformation in tri-axial compression test

The resistance to permanent deformation, in terms of the creep rate in tri-axial compression, f_c , shall be determined in accordance with EN 12697-25 using the conditions defined in prEN 13108-20:2013, D.7.

The resistance to permanent deformation of specimens shall comply with the category selected from Table 17.

The compaction of test specimens shall be selected from prEN 13108-20:2013, Table C.1.

The range between the upper and lower limits selected shall be 2 % based on compaction degree and 3 % based on void content.

The void content of the specimens shall be specified in accordance with prEN 13108-20:2013, D.2.

Table 17 — Maximum creep rate, $f_{c \max}$

Creep rate f_c $\mu\text{m/m}$	Category $f_{c \max}$
0,2	$f_{c \ 0,2}$
0,4	$f_{c \ 0,4}$
0,6	$f_{c \ 0,6}$
0,8	$f_{c \ 0,8}$
1,0	$f_{c \ 1,0}$
1,2	$f_{c \ 1,2}$
1,4	$f_{c \ 1,4}$
1,6	$f_{c \ 1,6}$
2	$f_{c \ 2}$
4	$f_{c \ 4}$
6	$f_{c \ 6}$
8	$f_{c \ 8}$
10	$f_{c \ 10}$
12	$f_{c \ 12}$
14	$f_{c \ 14}$
16	$f_{c \ 16}$
No requirement	$f_{c \ NR}$

5.3.6 Stiffness

The stiffness shall be determined in accordance with EN 12697-26 using the conditions defined in prEN 13108-20:2013, D.8.

The stiffness of specimens prepared shall comply with maximum and minimum categories selected from Tables 18 and 19 or shall be declared.

The compaction of test specimens shall be selected from prEN 13108-20:2013, Table C.1.

The range between the upper and lower limits selected shall be 2 % based on compaction degree and 3 % based on void content.

The void content of the specimens shall be specified in accordance with prEN 13108-20:2013, D.2.

Table 18 — Minimum stiffness, $S_{\min}$

Minimum stiffness MPa	Category $S_{\min}$
21 000	$S_{\min 21\ 000}$
17 000	$S_{\min 17\ 000}$
14 000	$S_{\min 14\ 000}$
11 000	$S_{\min 11\ 000}$
9 000	$S_{\min 9\ 000}$
7 000	$S_{\min 7\ 000}$
5 500	$S_{\min 5\ 500}$
4 500	$S_{\min 4\ 500}$
3 600	$S_{\min 3\ 600}$
2 800	$S_{\min 2\ 800}$
2 200	$S_{\min 2\ 200}$
1 800	$S_{\min 1\ 800}$
1 500	$S_{\min 1\ 500}$
Declared value	$S_{\min \text{ decl}}$
No requirement	$S_{\min \text{ NR}}$

Table 19 — Maximum stiffness, $S_{\max}$

Maximum stiffness MPa	Category $S_{\max}$
30 000	$S_{\max 30\ 000}$
25 000	$S_{\max 25\ 000}$
21 000	$S_{\max 21\ 000}$
17 000	$S_{\max 17\ 000}$
14 000	$S_{\max 14\ 000}$
11 000	$S_{\max 11\ 000}$
9 000	$S_{\max 9\ 000}$
7 000	$S_{\max 7\ 000}$
Declared value	$S_{\max \text{ decl}}$
No requirement	$S_{\max \text{ NR}}$

5.3.7 Resistance to fatigue

The resistance to fatigue shall be determined in accordance with EN 12697-24 using the conditions defined in prEN 13108-20:2013-...., D.9.

The minimum resistance to fatigue of specimens shall comply with the categories selected from Table 20 or shall be declared.

The compaction of test specimens shall be selected from prEN 13108-20:2013, Table C.1.

The range between the upper and lower limits selected shall be 2 % based on compaction degree and 3 % based on void content.

The void content of the specimens shall be specified in accordance with prEN 13108-20:2013, D.2.

Table 20 — Minimum resistance to fatigue, ε_6

Minimum resistance to fatigue microstrain	Category ε_6
310	ε_{6-310}
260	ε_{6-260}
220	ε_{6-220}
190	ε_{6-190}
160	ε_{6-160}
130	ε_{6-130}
115	ε_{6-115}
100	ε_{6-100}
90	ε_{6-90}
80	ε_{6-80}
70	ε_{6-70}
60	ε_{6-60}
50	ε_{6-50}
Declared value	ε_{6-decl}
No requirement	ε_{6-NR}

5.3.8 Mixture Saturation Ageing Tensile Stiffness conditioning test (SATS Durability Index)

The resistance Mixture SATS Durability Index shall be determined in accordance with EN 12697-45 using the conditions defined in prEN 13108-20:2013, D.17.

The Mixture SATS Durability Index of test specimens shall comply with the category selected from Table 21 or shall be declared.

The void content of the specimens shall be specified in accordance with prEN 13108-20:2013, D.2.

Table 21 — Mixture SATS Durability Index, MDI

Minimum SATS Durability Index %	Category MDI
100	MDI_{100}
85	MDI_{85}
70	MDI_{70}
55	MDI_{55}
Declared value	MDI_{Decl}
No requirement	MDI_{NR}

5.3.9 Low temperature properties

The low temperature properties shall be determined in accordance with EN 12697-46 using the conditions defined in prEN 13108-20:2013, D.18.

The low temperature properties of specimens, shall comply with the category selected from Table 22.

The compaction of test samples shall be selected from prEN 13108-20:2013, Table C.1.

The range between the upper and lower limits selected shall be 2 % based on compaction degree and 3 % based on void content.

The void content of the specimens shall be specified in accordance with prEN 13108-20:2013, D.2.

Table 22 — Maximum failure temperature, $TSRST_{max}$

Maximum failure temperature °C	Category $TSRST_{max}$
–15,0	$TSRST_{max 15,0}$
–17,5	$TSRST_{max 17,5}$
–20,0	$TSRST_{max 20,0}$
–22,5	$TSRST_{max 22,5}$
–25,0	$TSRST_{max 25,0}$
–27,5	$TSRST_{max 27,5}$
–30,0	$TSRST_{max 300}$
No requirement	$TSRST_{max NR}$

5.3.10 Fracture toughness

The fracture toughness properties shall be determined in accordance with EN 12697-44 using the conditions defined in prEN 13108-20:2013, D.19.

The fracture toughness properties of specimens shall comply with the category selected from Table 23.

The compaction of test samples shall be selected from prEN 13108-20:2013, Table C1.

The range between the upper and lower limits selected shall be 2 % based on compaction degree and 3 % based on void content.

The void content of the specimens shall be specified in accordance with prEN 13108-20:2013, D.2.

Table 23 — Fracture toughness, K_{Ic}

Maximum value N/mm ^{1,5}	Category K_{Ic}
10	$K_{Ic} 10$
15	$K_{Ic} 15$
20	$K_{Ic} 20$
25	$K_{Ic} 25$
30	$K_{Ic} 30$
35	$K_{Ic} 35$
40	$K_{Ic} 40$
45	$K_{Ic} 45$
50	$K_{Ic} 50$
55	$K_{Ic} 55$
No requirement	$K_{Ic} NR$

5.3.11 Friction after polishing

NOTE Friction after polishing is regarded as a property with direct relation to the development of friction in the road with time. As insufficient experience with the test method is available, it was considered to be too early to introduce this property as a mandatory requirement. In order to gain experience in Europe with this test method it is advised to investigate on a voluntary basis data obtained using this test (see Annex B).

5.3.12 Coating and homogeneity

The material when discharged from the mixer shall be homogenous in appearance with the aggregate completely coated with binder, and there shall be no evidence of balling of fine aggregate.

5.3.13 Reaction to fire

Where subject to regulation, the manufacturer shall declare the reaction to fire class according to EN 13501-1:2010, Table 2, according to the test method EN ISO 11925-2.

5.3.14 Marshall values for application on airfields

The Marshall values shall be determined according to EN 12697-34 using the conditions defined in prEN 13108-20:2013, D.10.

The Marshall Stability, the Marshall Flow and the Marshall Quotient, shall comply with the categories selected from Tables 23, 24, 25 and 26 respectively. For Marshall flow a minimum and maximum value may be selected, the range shall be at least 2.

Table 23 — Minimum Marshall Stability, $S_{\min}$

Minimum Marshall stability kN	Category $S_{\min}$
12,5	$S_{\min 12,5}$
10,0	$S_{\min 10,0}$
7,5	$S_{\min 7,5}$
5,0	$S_{\min 5,0}$
2,5	$S_{\min 2,5}$
No requirement	$S_{\min NR}$

Table 24 — Maximum Marshall Stability, $P_{\max}$

Maximum Marshall stability kN	Category $P_{\max}$
15,0	$P_{\max 15,0}$
12,5	$P_{\max 12,5}$
10,0	$P_{\max 10,0}$
7,5	$P_{\max 7,5}$
No requirement	$P_{\max NR}$

Table 25 — Marshall Flow, F

Marshall flow mm	Category F
1,0	$F_{1,0}$
1,5	$F_{1,5}$
2,0	$F_{2,0}$
3,0	$F_{3,0}$
4,0	$F_{4,0}$
5,0	$F_{5,0}$
8,0	$F_{8,0}$
No requirement	F_{NR}

Table 26 — Minimum Marshall Quotient, $Q_{\min}$

Minimum Marshall quotient kN/mm	Category $Q_{\min}$
4,0	$Q_{\min 4,0}$
3,5	$Q_{\min 3,5}$
3,0	$Q_{\min 3,0}$
2,5	$Q_{\min 2,5}$
2,0	$Q_{\min 2,0}$
1,5	$Q_{\min 1,5}$
1,0	$Q_{\min 1,0}$
No requirement	$Q_{\min NR}$

5.3.15 Resistance to fuel for application on airfields

The resistance to fuel shall be determined according EN 12697-43 using the conditions defined in prEN 13108-20:2013, Annex D.11.

The resistance to fuel of specimens shall comply with the category selected from Table 27.

The compaction of test specimens shall be selected from prEN 13108-20:2013-..., Table C.1.

The range between the upper and lower limits selected shall be 2 % based on compaction degree and 3 % based on void content.

Table 27 — Resistance to fuel

Resistance to fuel %	Category
1	Ci_1
2	Ci_2
3	Ci_3
4	Ci_4
5	Ci_5
6	Ci_6
7	Ci_7
8	Ci_8
No requirement	Ci_{NR}

5.3.16 Resistance to de-icing fluid for application on airfields

The resistance to de-icing fluids shall be determined according EN 12697-41 using the conditions defined in prEN 13108-20:2013, D.12.

The resistance to de-icing fluid of specimens shall comply with the category selected from Table 28.

The compaction of test specimens shall be selected from prEN 13108-20:2013-..., Table C.1.

The range between the upper and lower limits selected shall be 2 % based on compaction degree and 3 % based on void content.

Table 28 — Retained strength, β

Minimum retained strength %	Category β
100	β_{100}
85	β_{85}
70	β_{70}
55	β_{55}
No requirement	β_{NR}

5.4 Temperature of the mixture

The maximum temperature requirements apply at any place in the plant and shall be declared.

The minimum temperature of the mixture at delivery shall be declared.

Depending on local conditions and for specific application the minimum temperature, measured according to EN 12697-13, may be specified.

When using paving grade binder, the maximum temperature, measured according to EN 12697-13, shall not exceed the limits given in Table 29.

Table 29 — Maximum temperature of the mixture

Paving grade of binder	Maximum temperature °C
20/30	200
30/45	195
35/50, 40/60	190
50/70, 70/100	180
100/150, 160/220	170
250/330, 330/430	160

When using modified bitumen or hard grade bitumen, additives or premix bitumen, different temperatures may be applicable. These shall then be documented and declared.

5.5 Dangerous substances

When required, products covered by this standard shall comply with relevant regulations on dangerous substances in force in the intended place of use.

In the absence of International or European test methods, manufacturers shall verify and declare the release of dangerous substances in accordance with provisions applicable in the intended place of use of the product.

NOTE An informative database of European and national regulations on dangerous substances is available at the Construction web site on EUROPA (accessed through <http://ec.europa.eu/enterprise/construction/cpd-ds>).

5.6 Conflicting specifications

The overall quality of an Asphalt Concrete mixture can be covered by different combinations of requirements. The selection of requirements and the appropriate values shall be selected in such a way that conflicting specification is prevented.

To prevent this conflicting specification of mixtures the following combinations of requirements are not permissible:

- Requirements for Marshall values (see 5.3.14), for voids filled with bitumen (see 5.3.2.2), for voids in mineral aggregate (see 5.3.2.3) and for void content at 10 gyrations (see 5.3.2.4) shall not be combined with requirements for resistance to permanent deformation (see 5.3.5);
- requirements for binder content with a percentage above 3,0 shall not be combined with requirements for fatigue (5.3.7), stiffness (5.3.6) and Resistance to permanent deformation in tri-axial compression test (5.3.5.2);
- requirements for the stiffness properties of filler (4.3.4) and the angularity of fine sand (4.3.2) shall not be combined with requirements for stiffness, fatigue and resistance to permanent deformation in tri-axial compression test (5.3.5.2);
- requirements for resistance to permanent deformation for rut depth and rut rate in the wheel tracking (5.3.5.1.) shall not be combined with resistance to permanent deformation in tri-axial compression test (5.3.5.2).

6 Assessment and verification of constancy of performance — AVCP

The compliance of Asphalt Concrete with the requirements of this European Standard and with the performances declared by the manufacturer in the Declaration of performance (DoP) shall be demonstrated by

- determination of the product type in accordance with prEN 13108-20;
- factory production control by the manufacturer, including product assessment in accordance with prEN 13108-21.

The manufacturer shall always retain the overall control and shall have the necessary means to take responsibility for the conformity of the product with its declared performance(s).

For the purpose of Type Testing, Asphalt Concrete may be grouped into families as described in prEN 13108-20 where it is considered that the selected property or properties is or are common to all the mixtures within that family.

7 Identification

The delivery ticket shall contain at least the following information relating to identification:

- manufacturer and mixing plant;
- mix identification code;

— designation of the mixture:

AC	<i>D</i>	surf/base/bin	binder
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where

AC is Asphalt Concrete;

D is the upper sieve size;

surf is the surface course;

base is the base course;

bin is the binder course;

binder is the designation of binder category in the final product.

EXAMPLE AC 16 surf 70/100

- Asphalt Concrete with maximum aggregate size 16 mm for surface course with penetration bitumen 70/100.
- how to obtain the full details demonstrating conformity with this European Standard;
- details of any additives (see 4.1 and 4.5);
- details of compliance with 5.3.13, 5.3.14 and 5.3.15 where requested for specific use on airfields specifications.

NOTE Information concerning regulatory marking accompanies the product (For CE marking and labelling see ZA.3), but characteristics which are not necessarily part of regulatory marking, for example, special requirements for airfields could be made available by alternative means i.e. by placement on the CE mark certificate in a clearly identifiable area separated from the CE mark regulatory data or retained by the dispatching depot if agreed with the client.

Annex A (normative)

Calculations of the penetration or the softening point of the binder of a mixture when reclaimed asphalt is used

A.1 General

These calculations shall only be applied when paving grade bitumen has been used in the reclaimed asphalt and will be used as added binder.

A.2 Calculation of the penetration of the binder of a mixture

Use the following calculation:

$$a \lg pen_1 + b \lg pen_2 = (a + b) \lg pen_{mix}$$

where

pen_{mix} is the calculated penetration of the binder in the mixture containing reclaimed asphalt;

pen_1 is the penetration of the binder recovered from the reclaimed asphalt;

pen_2 is the penetration of the added binder;

a and b are the portions by mass of the binder from the reclaimed asphalt (a) and from the added binder (b) in the mixture; $a + b = 1$.

EXAMPLE $pen_1 = 20$; $pen_2 = 90$; $a = 0,25$ and $b = 0,75$
 $0,25 \lg 20 + 0,75 \lg 90 = \lg pen_{mix}$
 $\lg pen_{mix} = 1,79094$; therefore $pen_{mix} = 62$

The recovery of binder from mixtures for testing shall be executed according to EN 12697-3 or EN 12697-4.

The penetrations of the added binder and the recovered binder shall be determined according to EN 1426.

A.3 Calculation of the softening point of the binder of a mixture

Use the following calculation:

$$T_{R\&B\ mix} = a \times T_{R\&B\ 1} + b \times T_{R\&B\ 2}$$

where

$T_{R\&B\ mix}$ is the calculated softening point of the binder in the mixture containing reclaimed asphalt;

$T_{R\&B\ 1}$ is the softening point of the binder recovered from the reclaimed asphalt;

$T_{R\&B\ 2}$ is the softening point of the added binder;

a and b are the portions by mass of binder from the reclaimed asphalt (a) and from the added binder (b) in the mixture; $a + b = 1$.

EXAMPLE $T_{R\&B\ 1} = 62\ ^\circ\text{C}$; $T_{R\&B\ 2} = 48\ ^\circ\text{C}$; $a = 0,25$ and $b = 0,75$
 $T_{R\&B\ mix} = 0,25 \times 62 + 0,75 \times 48 = 51,5\ ^\circ\text{C}$

The softening points of the added binder and the recovered binder shall be determined according to EN 1427.

Annex B (informative)

Friction after polishing

Friction after polishing is regarded as a property with direct relation to the development of friction in the road in time. As insufficient experience with the test method is available, it was considered to be too early to introduce this property as a mandatory requirement. In order to gain experience in Europe with this test method it is advised to investigate on voluntary basis data with this test.

Below information on sample preparation is given.

The compaction of test specimens should be selected from prEN 13108-20:2013, Table C.1.

The range between the upper and lower limits selected should be 2 % based on compaction degree and 3 % based on void content.

The void content of the specimens should be specified in accordance with prEN 13108-20:2013, D.2.

The laboratory friction after polishing should be determined in accordance with EN 12697-49.

Based on experiences from different countries FAP values between 0,30 and 0,50 can be expected.

Annex ZA (informative)

Clauses of this European Standard addressing the provisions of the EU Construction Products Regulation

ZA.1 Scope and relevant characteristics

This European Standard has been prepared under Mandate M124 Road Construction Products given to CEN by the European Commission and the European Free Trade Association.

This annex deals with the CE marking of the Asphalt Concrete intended for the uses indicated in Table ZA.1 and shows the relevant clauses applicable.

This annex has the same scope as in Clause 1 of this standard related to the aspects covered by the mandate and is defined by Table ZA.1.

Table ZA.1 — Relevant clauses for Asphalt concrete product and intended use

Product: Asphalt concrete			
Intended use: For use on roads, airfields and other trafficked areas.			
Essential Characteristics	Clauses in this and other European Standard(s) related to essential characteristics	Regulatory classes	Notes
Adhesion of binder to aggregate	5.3.2.1 Maximum void content	—	Declared category
	5.3.2.1 Minimum void content	—	Declared category
	5.1/5.2.3 Target binder content	—	Declared value
	5.3.3 Water sensitivity	—	Declared category
	5.4 Temperature of the mixture	—	Declared maximum category and declared minimum value
Stiffness	5.1/5.2.2 Target Grading	—	Declared values
	5.1/5.2.3 Target binder content	—	Declared value
	5.3.2.1 Maximum void content	—	Declared maximum category
	5.3.2.1 Minimum void content	—	Declared minimum category
	5.4 Temperature of the mixture	—	Declared maximum category and declared minimum value
	5.3.6 Maximum stiffness	—	Declared maximum category or value
	5.3.6 Minimum stiffness	—	Declared minimum category or value

Product: Asphalt concrete Intended use: For use on roads, airfields and other trafficked areas.			
Essential Characteristics	Clauses in this and other European Standard(s) related to essential characteristics	Regulatory classes	Notes
Resistance to permanent deformation	5.1/5.2.2 Target Grading	—	Declared values
	5.1/5.2.3 Target binder content	—	Declared value
	5.3.2.1 Maximum void content	—	Declared maximum category
	5.3.2.1 Minimum void content	—	Declared minimum category
	5.3.2.2 Maximum Voids filled with bitumen	—	Declared maximum category
	5.3.2.2 Minimum voids filled with bitumen	—	Declared minimum category
	5.3.2.3 Minimum voids in mineral aggregate	—	Declared category
	5.4 Temperature of the mixture	—	Declared maximum category and declared minimum value
	5.3.14 Minimum Marshall stability for application on airfields	—	Declared minimum category
	Maximum Marshall stability for application on airfields		Declared maximum category
Resistance to fatigue	Minimum Marshall flow for application on airfields		Declared minimum category
	Maximum Marshall flow for application on airfields		Declared maximum category
	Minimum Marshall quotient for application on airfields		Declared minimum category
	Maximum Marshall quotient for application on airfields		Declared maximum category
	5.3.5 Resistance to permanent deformation	—	Declared maximum categories
	5.3.2.1 Maximum void content	—	Declared maximum category
	5.3.2.1 Minimum void content	—	Declared minimum category
	5.4 Temperature of the mixture	—	Declared minimum category or declared minimum value
	5.1/5.2.3 Target binder content	—	Declared value
	5.3.7 Resistance to fatigue	—	Declared minimum value or category

Product: Asphalt concrete			
Intended use: For use on roads, airfields and other trafficked areas.			
Essential Characteristics	Clauses in this and other European Standard(s) related to essential characteristics	Regulatory classes	Notes
Skid resistance	5.1/5.2.2 Target Grading	–	Declared value
	5.1/5.2.3 Target binder content	–	Declared value
	5.3.2.1 Maximum void content	–	Declared maximum category
	5.3.2.1 Minimum void content	–	Declared minimum category
Resistance to abrasion	5.1/5.2.2 Target Grading	–	Declared value
	5.1/5.2.3 Target binder content	–	Declared value
	5.3.4 Resistance to abrasion by studded tires	–	Declared maximum category
Reaction to fire ^a	5.3.13 Fire class	A1 _{fl} to F _{fl}	Declared class
Dangerous substances	5.5 Dangerous substances		
Durability of the above characteristics against ageing, weathering, oxidation, wear, ravelling, chemicals, wear of studded tyres, stripping, ... as relevant	All above mentioned requirements are related to durability.		
	5.3.8 Mixture SATS Durability Index	–	Minimum category
	5.3.9 Low temperature property	–	Maximum category
	5.3.10 Fracture toughness	–	Maximum category
	5.3.15 Resistance to fuel for application on airfields	–	Declared category
	5.3.16 Resistance to de-icing fluids for application on airfields	–	Declared minimum category
^a Relevant only for Asphalt Concrete intended for uses subject to reaction to fire regulations.			

The declaration of the product performance related to certain essential characteristics is not required in those Member States (MS) where there are no regulatory requirements on these essential characteristics for the intended use of the product.

In this case, manufacturers placing their products on the market of these MS are not obliged to determine nor declare the performance of their products with regard to these essential characteristics and the option “No performance determined” (NPD) in the information accompanying the CE marking and in the declaration of performance (see ZA.3) may be used for those essential characteristics.

ZA.2 Procedure for AVCP of Asphalt Concrete

ZA.2.1 System(s) of AVCP

The AVCP systems of asphalt concrete indicated in Table ZA.1, established by EC Decision 1998/601/EC (OJ L 287, 24.1.1998, p. 41) amended by the Commission decision 2001/596/EC of 8 January 2001 (OJ L 209; p. 33) are shown in Table ZA.2 for the indicated intended uses and relevant levels or classes of performance.

Table ZA.2 — Systems of AVCP

Products	Intended uses	Levels or classes of performance	AVCP systems
Bituminous mixtures	For road construction and surface treatment of roads	–	2+
Bituminous mixtures	For uses subject to reaction to fire regulations	A1 _{fl} ⁽¹⁾ , A2 _{fl} ⁽¹⁾ , B _{fl} ⁽¹⁾ , C _{fl} ⁽¹⁾	1
		A1 _{fl} ⁽²⁾ , A2 _{fl} ⁽²⁾ , B _{fl} ⁽²⁾ , C _{fl} ⁽²⁾ , D _{fl} , E _{fl}	3
		(A1 _{fl} to E _{fl}) ⁽³⁾ , F _{fl}	4
System 1: See Regulation (EU) No. 305/2011 (CPR) Annex V, 1.2.			
System 2+: See Regulation (EU) No. 305/2011 (CPR) Annex V, 1.3 including certification of the factory production control by a notified production control certification body on the basis of initial inspection of the manufacturing plant and of factory production control as well as of continuous surveillance, assessment and evaluation of factory production control.			
System 3: See Regulation (EU) No. 305/2011 (CPR) Annex V, 1.4.			
System 4: See Regulation (EU) No. 305/2011 (CPR) Annex V, 1.5.			
(1) Products/materials for which a clearly identifiable stage in the production process results in an improvement of the reaction to fire classification (e.g. an addition of fire retardants or a limiting of organic material).			
(2) Products/materials not covered by footnote 1.			
(3) Products/materials that do not require to be tested for reaction to fire (e.g. Products/materials of Classes A1 according to Commission Decision 96/603/EC).			

The AVCP of the Asphalt Concrete in Table ZA.1 shall be according to the AVCP procedures indicated in Tables ZA.3.1 to ZA.3.3 resulting from application of the clauses of this or other European Standard indicated therein. The content of tasks of the notified body shall be limited to those essential characteristics as provided for, if any, in Annex III of the relevant mandate and to those that the manufacturer intends to declare.

Table ZA.3.1 — Assignment of AVCP tasks for asphalt concrete under system 2+ and subject to reaction to fire under system 1

Tasks		Content of the task	AVCP clauses to apply
Tasks for the manufacturer	Factory production control (FPC)	Parameters related to essential characteristics of Table ZA.1 relevant for the intended use which are declared.	prEN 13108-21
	Determination of the product-type on the basis of type testing (including sampling), type calculation, tabulated values or descriptive documentation of the product.	Parameters related to essential characteristics of Table ZA.1 relevant for the intended use which are declared except reaction to fire.	prEN 13108-20
	Further testing of samples taken at factory according to the prescribed test plan.	Essential characteristics of Table ZA.1 relevant for the intended use which are declared.	prEN 13108-21
Tasks for the notified product certification body	determination of the product type on the basis of type testing (including sampling), type calculation, tabulated values or descriptive documentation of the product.	Reaction to fire	prEN 13108-20
	Initial inspection of manufacturing plant and of FPC	Parameters related to essential characteristic of Table ZA.1, relevant for the intended use which is declared, namely Reaction to fire. Documentation of the FPC.	prEN 13108-21
	Continuous surveillance, assessment and evaluation of FPC	Parameters related to essential characteristic of Table ZA.1, relevant for the intended use which is declared, namely Reaction to fire. Documentation of FPC.	prEN 13108-21
Tasks for the notified production control certification body	Initial inspection of the manufacturing plant and of FPC	Parameters related to essential characteristics of Table ZA.1, relevant for the intended use which is declared, except reaction to fire. Documentation of the FPC.	prEN 13108-21
	Continuous surveillance, assessment and evaluation of FPC	Parameters related to essential characteristics of Table ZA.1, relevant for the intended use which is declared, except reaction to fire. Documentation of the FPC.	prEN 13108-21

Table ZA.3.2 — Assignment of AVCP tasks for Asphalt concrete under system 2+ and subject to reaction to fire under system 3

Tasks		Content of the task	AVCP clauses to apply
Tasks for the manufacturer	Factory production control (FPC)	Parameters related to essential characteristics of Table ZA.1 relevant for the intended use which are declared.	prEN 13108-21
	Determination of the product-type on the basis of type testing (including sampling), type calculation, tabulated values or descriptive documentation of the product.	Parameters related to essential characteristics of Table ZA.1 relevant for the intended use which are declared except reaction to fire.	prEN 13108-20
	Further testing of samples taken at factory according to the prescribed test plan.	Essential characteristics of Table ZA.1 relevant for the intended use which are declared.	prEN 13108-21
Tasks for a notified testing laboratory	Determination of the product-type on the basis of type testing (based on sampling carried out by the manufacturer), type calculation, tabulated values or descriptive documentation of the product.	Reaction to fire	prEN 13108-20
Tasks for the notified production control certification body	Initial inspection of the manufacturing plant and of FPC	Parameters related to essential characteristics of Table ZA.1, relevant for the intended use which is declared. Documentation of the FPC.	prEN 13108-21
	Continuous surveillance, assessment and evaluation of FPC	Parameters related to essential characteristics of Table ZA.1, relevant for the intended use which is declared, namely. Documentation of the FPC.	prEN 13108-21: 2013 Annex B

Table ZA.3.3 — Assignment of AVCP tasks for Asphalt Concrete under system 2+ and subject to Reaction to fire under system 4

Tasks		Content of the task	AVCP clauses to apply
Tasks for the manufacturer	Factory production control (FPC)	Parameters related to essential characteristics of Table ZA.1 relevant for the intended use which is declared.	prEN 13108-21
	Determination of the product-type on the basis of type testing (including sampling), type calculation, tabulated values or descriptive documentation of the product.	Parameters related to essential characteristics of Table ZA.1 relevant for the intended use which is declared.	prEN 13108-20
	Further testing of samples taken at factory according to the prescribed test plan.	Essential characteristics of Table ZA.1 relevant for the intended use which is declared.	prEN 13108-21
Tasks for the notified production control certification body	Initial inspection of the manufacturing plant and of FPC.	Parameters related to essential characteristics of Table ZA.1, relevant for the intended use which is declared. Documentation of the FPC.	prEN 13108-21
	Continuous surveillance, assessment and evaluation of FPC.	Parameters related to essential characteristics of Table ZA.1, relevant for the intended use which is declared. Documentation of the FPC.	prEN 13108-21

ZA.2.2 Declaration of performance (DoP)

ZA.2.2.1 General

The manufacturer draws up the DoP and affixes the CE marking on the basis of the different AVCP systems set out in Annex V of the Regulation (EU) No 305/2011:

In case of products under system 1

- the factory production control and further testing of samples taken at the factory according to the prescribed test plan, carried out by the manufacturer; and
- the certificate of constancy of performance issued by the notified product certification body on the basis of determination of the product type on the basis of type testing (including sampling), type calculation, tabulated values or descriptive documentation of the product; initial inspection of the manufacturing plant and of factory production control and continuous surveillance, assessment and evaluation of factory production control.

In case of products under system 2+

- the determination of the product-type on the basis of type testing (including sampling), type calculation, tabulated values or descriptive documentation of the product; the factory production control and the testing of samples taken at the factory according to the prescribed test plan, carried out by the manufacturer; and

- the certificate of conformity of the factory production control, issued by the notified production control certification body on the basis of:
 - initial inspection of the manufacturing plant and of factory production control; and
 - continuous surveillance, assessment and evaluation of factory production control.

In case of products under system 3

- the factory production control carried out by the manufacturer; and
- the determination of the product-type on the basis of type testing (based on sampling carried out by the manufacturer), type calculation, tabulated values or descriptive documentation of the product, carried out by the notified testing laboratory.

In case of products under system 4

- the factory production control carried out by the manufacturer;
- the determination by the manufacturer of the product-type on the basis of type testing, type calculation, tabulated values or descriptive documentation of the product.

ZA.2.2.2 Content

The model of the DoP is provided in Annex III of the Regulation (EU) No 305/2011.

According to this Regulation, the DoP shall contain, in particular, the following information:

- the reference of the product-type for which the declaration of performance has been drawn up;
- the AVCP system or systems of the construction product, as set out in Annex V of the CPR;
- the reference number and date of issue of the harmonised standard which has been used for the assessment of each essential characteristic;
- where applicable, the reference number of the Specific Technical Documentation used and the requirements with which the manufacturer claims the product complies.

The DoP shall in addition contain:

- a) the intended use or uses for the construction product, in accordance with the applicable harmonised technical specification;
- b) the list of essential characteristics, as determined in the harmonised technical specification for the declared intended use or uses;
- c) the performance of at least one of the essential characteristics of the construction product, relevant for the declared intended use or uses;
- d) where applicable, the performance of the construction product, by levels or classes, or in a description, if necessary based on a calculation in relation to its essential characteristics determined in accordance with the Commission determination regarding those essential characteristics for which the manufacturer shall declare the performance of the product when it is placed on the market or the Commission determination regarding threshold levels for the performance in relation to the essential characteristics to be declared.
- e) the performance of those essential characteristics of the construction product which are related to the intended use or uses, taking into consideration the provisions in relation to the intended use or uses where the manufacturer intends the product to be made available on the market;
- f) for the listed essential characteristics for which no performance is declared, the letters “NPD” (No Performance Determined).

Regarding the supply of the DoP, article 7 of the Regulation (EU) No 305/2011 applies.

The information referred to in Article 31 or, as the case may be, in Article 33 of Regulation (EC) No 1907/2006, (REACH) shall be provided together with the DoP.

ZA.2.2.3 Example of DoP

The following gives an example of a filled-in DoP for vertical air/flue terminals

DECLARATION OF PERFORMANCE No. 001CPR2013-07-14

1. Unique identification code of the product-type:

**Asphalt Concrete
AC XV124**

2. Type, batch or serial number or any other element allowing identification of the construction product as required under Article 11(4):

**Asphalt Concrete
AC XV124 — 0045**

3. Intended use or uses of the construction product, in accordance with the applicable harmonised technical specification, as foreseen by the manufacturer:

For surface courses.

4. Name, registered trade name or registered trade mark and contact address of the manufacturer as required under Article 11(5):

**AnyCo SA,
PO Box 21
B-1050 Brussels, Belgium
Tel. +32987654321
Fax: +32123456789
E-mail: anyco.sa@provider.be**

5. Where applicable, name and contact address of the authorised representative whose mandate covers the tasks specified in Article 12(2):

**Delorme-Zwan
Anyco Ltd
Rue de la Standardisation
Paris
France
Tel. +33987654321
Fax: +33123456789
E-mail: anyone.ltd@provider.fr**

6. System or systems of assessment and verification of constancy of performance of the construction product as set out in CPR, Annex V:

System 2+ and system 1 (reaction to fire)

7. In case of the declaration of performance concerning a construction product covered by a harmonised standard: Notified factory production control certification body No. 5678 performed the initial inspection of the manufacturing plant and of factory production control and the continuous surveillance, assessment and evaluation of factory production control and issued the certificate of conformity of the factory production control.

8. Declared performance.

Essential characteristics	Performance	Harmonised technical specification EN 13108-1:2013																		
1. Adhesion of binder to aggregate 2. Stiffness 3. Resistance to permanent deformation 4. Resistance to fatigue 5. Skid resistance 6. Resistance to abrasion 7. Reaction to Fire 8. Dangerous substances 9. Durability																				
2, 3, 4, 5	Target grading passing sieve <table><tr><th>Sieve mm</th><th>Passing %</th></tr><tr><td>22,4</td><td>100</td></tr><tr><td>16</td><td>99</td></tr><tr><td>11,2</td><td>88</td></tr><tr><td>5,6</td><td>65</td></tr><tr><td>2</td><td>43</td></tr><tr><td>0,5</td><td>32</td></tr><tr><td>0,125</td><td>15</td></tr><tr><td>0,063</td><td>7,2</td></tr></table>	Sieve mm	Passing %	22,4	100	16	99	11,2	88	5,6	65	2	43	0,5	32	0,125	15	0,063	7,2	
Sieve mm	Passing %																			
22,4	100																			
16	99																			
11,2	88																			
5,6	65																			
2	43																			
0,5	32																			
0,125	15																			
0,063	7,2																			
1, 2, 3, 4, 5, 6	Target binder content 5,7 %																			
1, 2, 4, 5	Minimum and maximum void content $V_{\min 4,0}$; $V_{\max 8,0}$																			
1	Water sensitivity i/C_{80} .	EN 12697-12 Method B																		
1, 2, 3, 4	Maximum Temperature 180 °C Minimum temperature 135 °C																			
2	Stiffness Minimum Stiffness 9 000 MPa Maximum Stiffness NPD	EN 12697-26 Annex B																		
3	Maximum creep rate f_c 1,0	EN 12697-25 Method B																		
4	Resistance to fatigue ε_{6-130}	EN 12697-24 Annex D																		
6	Resistance to abrasion Abr_{B36}	EN 12697-16 Method B																		
7	F_{fl}																			
8	X: Less than 60 mg/m ²																			
9	Mixture SATS durability index NPD																			
9	Low temperature property $TSRST_{\max-27,5}$																			

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ZA.3 CE marking and labelling

The CE marking symbol shall be in accordance with the general principles set out in Article 30 of Regulation (EC) No 765/2008 and shall be affixed visibly, legibly and indelibly:

- to the packaging

or

- to the accompanying documents.

The CE marking shall be followed by:

- the last two digits of the year in which it was first affixed,
- the name and the registered address of the manufacturer, or the identifying mark allowing identification of the name and address of the manufacturer easily and without any ambiguity,
- the unique identification code of the product-type,
- the reference number of the declaration of performance [see example of DoP],
- the level or class of the performance declared,
- the dated reference to the harmonised technical specification applied,
- the identification number of the notified body,
- the intended use as laid down in the harmonised technical specification applied.

The CE marking shall be affixed before the construction product is placed on the market. It may be followed by a pictogram or any other mark notably indicating a special risk or use.

Figure ZA.1 gives an example of the information related to products subject to AVCP under system 2+ to be given on the accompanying documents.

																			
0123																			
AnyCo Ltd, PO Box 21, B-1050, Brussels, Belgium																			
13																			
001-CPR-2013/07/14																			
EN 13108-1:2013 Asphalt concrete AC XV124 Intended to be used for surface courses																			
Void content:																			
max.	8,0 %																		
min.	4,0 %																		
Maximum voids filled with bitumen	$VFB_{\max}$ 80																		
Minimum voids filled with bitumen	$VFB_{\min}$ 60																		
Voids in mineral aggregate	$VMA_{\min}$ 14																		
Water sensitivity	ilC_{80}																		
Resistance to abrasion by studded tires	Abr_B 36																		
Reaction to fire	C_{fl}																		
Resistance to fuel	NPD																		
Resistance to de-icing fluid	NPD																		
Maximum Temperature of the mixture	180 °C																		
Minimum Temperature of the mixture	135 °C																		
Grading target composition																			
<table border="1" style="border-collapse: collapse; width: 100%;"> <thead> <tr> <th style="padding: 5px;">Sieve size mm</th> <th style="padding: 5px;">Passing %</th> </tr> </thead> <tbody> <tr><td style="text-align: center; padding: 5px;">22,4</td><td style="text-align: center; padding: 5px;">100</td></tr> <tr><td style="text-align: center; padding: 5px;">16</td><td style="text-align: center; padding: 5px;">99</td></tr> <tr><td style="text-align: center; padding: 5px;">11,2</td><td style="text-align: center; padding: 5px;">88</td></tr> <tr><td style="text-align: center; padding: 5px;">5,6</td><td style="text-align: center; padding: 5px;">65</td></tr> <tr><td style="text-align: center; padding: 5px;">2</td><td style="text-align: center; padding: 5px;">43</td></tr> <tr><td style="text-align: center; padding: 5px;">0,5</td><td style="text-align: center; padding: 5px;">32</td></tr> <tr><td style="text-align: center; padding: 5px;">0,125</td><td style="text-align: center; padding: 5px;">15</td></tr> <tr><td style="text-align: center; padding: 5px;">0,063</td><td style="text-align: center; padding: 5px;">7,2</td></tr> </tbody> </table>	Sieve size mm	Passing %	22,4	100	16	99	11,2	88	5,6	65	2	43	0,5	32	0,125	15	0,063	7,2	
Sieve size mm	Passing %																		
22,4	100																		
16	99																		
11,2	88																		
5,6	65																		
2	43																		
0,5	32																		
0,125	15																		
0,063	7,2																		
Target Binder content	5,7 %																		
Marshall values																			
Marshall stability	NPD																		
Marshall flow	NPD																		
Marshall quotient	NPD																		
Resistance to permanent deformation	f_c 1,0																		
Minimum stiffness	9 000 MPa																		
Maximum stiffness	NPD																		
Resistance to fatigue	ϵ_{6-130}																		
Dangerous substance X:	less than 60 mg/m ²																		

CE marking, consisting of the “CE”-symbol

Identification number of the product certification body

Name and the registered address of the manufacturer, or identifying mark

Last two digits of the year in which the marking was first affixed

Reference number of the DoP

No. of European standard applied, as referenced in OJEU

Unique identification code of the product-type

Intended use of the product as laid down in the European standard applied

Level or class of the performance declared

Figure ZA.1 — Example CE marking information of products under AVCP system 2+ affixed to the accompanying documents

Introduction

This draft standard is based on European discussions in which the UK has taken an active part. Your comments on this draft are welcome and will assist in the preparation of the consequent British Standard. Comment is particularly welcome on national, legislative or similar deviations that may be necessary.

Even if this draft standard is not approved by the UK, if it receives the necessary support in Europe, the UK will be obliged to publish the official English Language text unchanged as a British Standard and to withdraw any conflicting standard.

UK Vote

Please indicate whether you consider the UK should submit a negative (with reasons) or positive vote on this draft.

Submission of Comments

- The guidance given below is intended to ensure that all comments receive efficient and appropriate attention by the responsible BSI committee. **Annotated drafts are not acceptable and will be rejected.**
- All comments must be submitted, preferably electronically, to the Responsible Committee Secretary at the address given on the front cover. Comments should be compatible with version 6.0 or version 97 of Microsoft Word for Windows, if possible; otherwise comments in ASCII text format are acceptable. **Any comments not submitted electronically should still adhere to these format requirements.**
- All comments submitted should be presented as given in the example below. Further information on submitting comments and how to obtain a blank electronic version of a comment form are available from the BSI website at:
<http://drafts.bsigroup.com/>

Template for comments and secretariat observations

Date: xx/xx/20xx	Document: ISO/DIS xxxx
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1	2	(3)	4	5	(6)	(7)
MB	Clause No./ Subclause No./Annex (e.g. 3.1)	Paragraph/ Figure/ Table/Note	Type of comment	Comment (justification for change) by the MB	Proposed change by the MB	Secretariat observations on each comment submitted
	3.1	Definition 1	ed	Definition is ambiguous and needs clarifying.	Amend to read '...so that the mains connector to which no connection...'	
	6.4	Paragraph 2	te	The use of the UV photometer as an alternative cannot be supported as serious problems have been encountered in its use in the UK.	Delete reference to UV photometer.	